

Solution Requirements

HDI Predictor – Human Development Index Prediction & Recommendation Platform

Date	08/07/2026
Team ID	
Project Name	HDI Predictor – Human Development Index Prediction & Recommendation Platform
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

This project builds a web application that predicts a country's (or hypothetical scenario's) Human Development Index (HDI) tier — Very High, High, Medium, or Low — based on three key indicators: Life Expectancy, Education (Mean/Expected Years of Schooling), and GNI per capita (PPP). The system also generates actionable recommendations for improving the weakest indicator when the predicted tier is below Very High. Below are the comprehensive functional and non-functional requirements for the solution.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	• Users (Public Users, Researchers, Policymakers) access the platform without mandatory login for basic predictions. • Registered users (Researchers/Policymakers) log in via email + password with bcrypt hashing. • Optional Google OAuth 2.0 login for faster access. • Session managed via JWT tokens with a 24-hour expiry.	Medium
2	Authorization levels	• Guest: Can input indicators and view predicted HDI tier and score (no save/history). • Registered User (Researcher/Policymaker): Can save prediction history, download reports, and compare multiple countries. • Admin: Can manage/update the underlying ML model, datasets, and view usage analytics.	Medium
3	External interfaces	• Integration with UNDP Human Development Report Open Data API for reference/historical HDI values. • REST API layer (Flask/FastAPI or Node.js) connecting the frontend to the trained ML model (scikit-learn/joblib pickle).	High

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		Optional integration with World Bank Open Data API for supplementary GNI, education, and life-expectancy datasets.	
4	Transactions processing	- Core transaction: user submits 3 indicators (Life Expectancy, Mean/Expected Years of Schooling, GNI per capita PPP) and receives a predicted HDI score + tier in real time (<2s). - No monetary transactions; all processing is data input → model inference → result output. - Each prediction request/response is logged with a timestamp and unique request ID for traceability.	High
5	Reporting	- Downloadable PDF/CSV report per prediction showing input values, predicted HDI score, tier classification, and improvement recommendations. - Comparison dashboard allowing side-by-side view of 2+ countries or scenarios. - Admin analytics dashboard: number of predictions made, most common tier predicted, average indicator values submitted.	Medium
6	Business rules, etc.	- HDI Tier classification thresholds: Very High (HDI ≥ 0.800), High (0.700–0.799), Medium (0.550–0.699), Low (< 0.550), consistent with UNDP methodology. - If any indicator is missing or out of realistic bounds (e.g., life expectancy > 90, GNI negative), system must prompt the user with validation errors before prediction. - System must generate at least one actionable improvement recommendation for any tier below "Very High", targeting the weakest-performing indicator.	High
7	Compliance to laws or regulations	- GDPR-aligned handling of any personally identifiable data collected during registration (name, email). - No sensitive personal data beyond account credentials is stored; country-level statistical inputs are non-personal. - Data retention and deletion policy: registered users can request permanent deletion of saved prediction history.	Medium
8	Other	Interactive "What-if" sliders allowing users to adjust one indicator at a time and see the predicted HDI tier update live, to explore possible improvement paths.	Low

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
		• Educational tooltips/info-icons explaining each indicator and how it factors into HDI. • Multi-scenario save/compare feature for policymakers evaluating multiple intervention strategies.	

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	• The ML model must return the predicted HDI score/tier immediately after the user submits the three indicators. • Frontend interactive sliders must update visual feedback with minimal lag for a smooth "what-if" experience.	• Prediction response time < 1.5 seconds. • Slider-to-chart UI update < 300 ms. • Page load < 2 seconds on 4G.
2	Scalability	• The platform must support simultaneous use by students, researchers, and policymakers, including classroom/batch usage during workshops.	• Supports up to 5,000 concurrent users. • Horizontally scalable API layer (stateless, containerized).
3	Security & Data Privacy	• All data in transit must be encrypted; stored credentials must be hashed, not plain text. • Model files and datasets protected from unauthorized modification.	• TLS 1.2+ for all traffic. • Passwords hashed with bcrypt (cost factor ≥ 10). • Role-based access control on admin endpoints.
4	Reliability & Availability	• The prediction service must remain available for continuous research and policy-planning use, including graceful handling of model-service downtime.	• 99.5% uptime per month. • Fallback error message if ML service is unreachable, with automatic retry.
5	Usability & Accessibility	• The interface must be intuitive for non-technical policymakers as well as technical researchers, using plain-language labels for each indicator and clear visual tier indicators (color-coded: green/blue/orange/red).	• Complies with WCAG 2.1 AA standards. • Usability tested with non-technical users; task completion in < 3 clicks.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
6	Other	• Maintainability: model retraining pipeline should be swappable without frontend changes. • Portability: web app should be responsive across desktop, tablet, and mobile. • Extensibility: architecture should allow adding more HDI-related indicators (e.g., Gini index) in future versions.	• Model versioning via config file, no hard-coded model logic in frontend. • Responsive layout tested on screen widths from 360px to 1920px.

Reviewer's Evaluation Notes

As a project reviewer, this requirements set is evaluated against: (1) Coverage — does it capture the full user journey from data input to tiered prediction to improvement recommendation, for all three problem-statement scenarios (Very High, Medium, Low)? (2) Interactivity — do the FRs support user-adjustable "what-if" exploration rather than a single static prediction? (3) Explainability — does the system justify WHY a tier was predicted and WHAT to improve, not just output a label? (4) Feasibility — are the NFR targets (response time, uptime, accessibility) realistic for a classic web-based ML-integrated application? This document addresses all four criteria: FR items 1–8 cover authentication through business rules and recommendation logic, and NFR items 1–6 cover performance, scalability, security, reliability, usability, and maintainability.